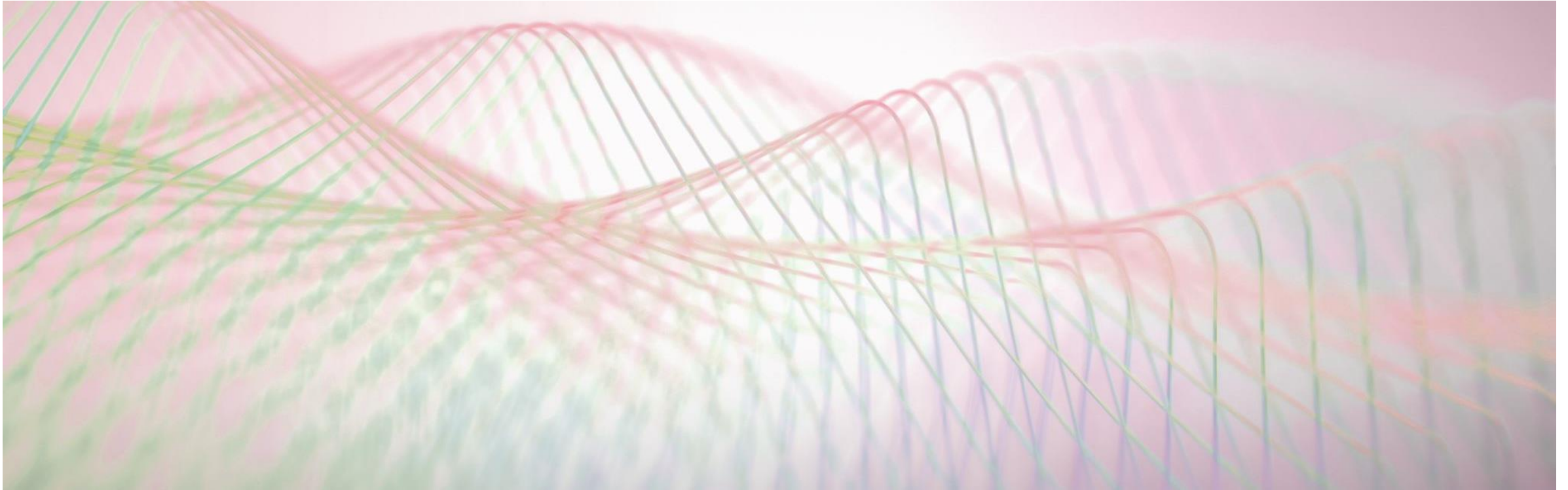




MATERIALS SCIENCE 2

ELECTRICAL AND THERMAL PROPERTIES OF MATERIALS

INTRODUCTION TO WEEK 12

DR ADIJAT 'WUMI INYANG (ADIJAT.INYANG@QMUL.AC.UK)

AIMS FOR THIS WEEK

01

Introduce the student to fundamental concepts of electrical and thermal properties of materials

02

Outline the process of conduction in terms of band and atomic bonding models

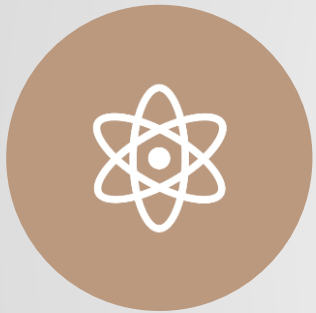
03

Describe the concept of semi-conduction and dielectric behaviour

04

Explain heat capacity, heat conduction and thermo-mechanical behaviour

LEARNING OUTCOMES



Describe the four possible electron band structures for solid materials



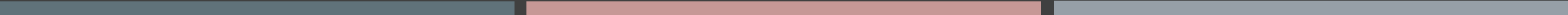
Describe electron events that produce free electrons/holes in materials



Perform calculations of electrical properties,



Identify application areas for semiconductor and dielectric materials



WEEK OUTLINE

Live stream of lecture

Monday

Live stream of lecture

Wednesday

Thursday

Live stream of lecture

SKILLS



EXPLAIN ELECTRICAL
PROPERTIES OF MATERIALS
USING ATOMIC APPROACH



DATA INTERPRETATION



LISTENING TECHNICAL
ENGLISH



PERFORM CALCULATIONS
AND INTERPRET RESULTS

QUESTIONS FOR THIS MODULE



WECOM WORK GROUPS: DURING
CLASS. THIS IS NOT MONITORED
OUTSIDE CLASS TIME



QMPLUS FORUM: ANY QUESTION
OUTSIDE CLASS TIME



EMAIL: PERSONAL ISSUES, NOT
RELATED TO HOMEWORK, CLASS
MATERIAL, MODULE CONTENT